

Malaysian Industrial Validation Report

Interpretable Machine Learning for Predicting GHS Chemical Hazard Classifications

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1. Purpose and scope

This report tests whether a machine-learning model trained on global GHS classification data performs adequately on the chemicals actually handled by Malaysian industry. Four sectors were examined - palm oil processing, rubber processing, petrochemicals and semiconductor manufacturing in Penang - together with the chemicals implicated in the Sungai Kim Kim incident at Pasir Gudang, Johor, in March 2019.

The best-performing model, **XGBoost**, was selected on a scaffold-split global test set of 24,352 compounds, where it achieved a mean AUC-ROC of 0.915 across the nine GHS hazard classes. The same model, the same descriptors and the same calibrated decision thresholds were applied unchanged here.

2. The Malaysian validation set

Sector	Compounds resolved	With GHS ground truth
Johor 2019 Emergency	12	12
Palm Oil Processing	8	7
Petrochemicals	9	9
Rubber Processing	7	7
Semiconductor (Penang)	8	8

Substitutions. Three entries in the proposal name materials rather than single molecules. A representative structure was used for each, and this is recorded so that no reader is misled: bleaching earth → Bentonite; fatty acids → Palmitic Acid; mercaptobenzothiazole → 2-Mercaptobenzothiazole; carbon black → Carbon; xylene → P-Xylene; xylene → P-Xylene.

3. Performance on Malaysian chemicals

GHS class	Meaning	n+	Acc.	Prec.	Recall	F1
GHS01	Explosive	0	1.00	0.00	0.00	0.00
GHS02	Flammable	28	0.88	0.87	0.96	0.92
GHS03	Oxidiser	2	1.00	1.00	1.00	1.00
GHS04	Compressed gas	10	0.95	0.83	1.00	0.91
GHS05	Corrosive	14	0.86	0.83	0.71	0.77
GHS06	Acute toxicity	16	0.79	0.65	0.94	0.77
GHS07	Irritant / harmful	37	0.86	0.94	0.89	0.92
GHS08	Serious health hazard	36	0.77	0.86	0.86	0.86
GHS09	Environmental hazard	25	0.49	0.80	0.16	0.27

By sector.

Sector	n	Label accuracy	Hazard recall
Johor 2019 Emergency	12	0.907	0.836
Palm Oil Processing	7	0.905	0.900
Petrochemicals	9	0.876	0.839
Rubber Processing	7	0.762	0.708
Semiconductor (Penang)	8	0.736	0.658

4. Sungai Kim Kim, Pasir Gudang, March 2019

On 7 March 2019 chemical waste was illegally discharged into the Sungai Kim Kim river at Pasir Gudang, Johor. Vapour rising from the river affected more than 2,500 people, the majority of them schoolchildren, and 111 schools were closed. Acrylonitrile and acrolein were identified by the Ministry of Health as the principal agents. The table below shows what this framework would have predicted for each of the chemicals named in the official and peer-reviewed accounts of the incident.

Compound	Role in the incident	Hazards flagged
acrylonitrile	principal agent identified by the Ministry o	GHS02, GHS05, GHS06, GHS07, GHS08, GHS09
acrolein	principal agent; a severe respiratory irrita	GHS02, GHS05, GHS06, GHS07, GHS08
benzene	volatile aromatic detected in air and water	GHS02, GHS07, GHS08
toluene	volatile aromatic detected in air samples	GHS02, GHS07, GHS08
xylene	volatile aromatic detected in air samples	GHS02, GHS07, GHS08
ethylbenzene	volatile aromatic detected in air samples	GHS02, GHS07, GHS08
hydrogen sulfide	released from the anaerobic river sediment	GHS02, GHS04, GHS06, GHS07, GHS08
methane	released from the anaerobic river sediment	GHS02, GHS04, GHS08
methyl mercaptan	odorous sulfur compound reported at the site	GHS02, GHS04, GHS06, GHS07, GHS08
d-limonene	solvent component reported in the waste mixt	GHS02, GHS07, GHS08, GHS09
hydrogen chloride	acid gas reported in the waste mixture	GHS04, GHS06, GHS07, GHS08
styrene	monomer reported among the solvent residues	GHS02, GHS07, GHS08

Interpretation. Across the incident mixture the framework most frequently flagged GHS08 (Serious health hazard, 12/12 compounds), GHS02 (Flammable, 11/12 compounds), GHS07 (Irritant / harmful, 11/12 compounds), GHS06 (Acute toxicity, 5/12 compounds). A screening tool producing this profile from nothing but the chemical structures would have supported an early decision to treat the discharge as an acute inhalation hazard rather than as ordinary waste.

5. Recommendations for DOSH adoption

1. Use as a triage tool, not a replacement for testing. The framework assigns a hazard profile from structure alone in under a second, which makes it suitable for prioritising which of thousands of registered chemicals should be tested first. It cannot substitute for laboratory determination under the CLASS Regulations 2013.
2. Operate at the safety-first threshold. Each hazard class has a calibrated threshold that catches at least 90% of true hazards, at the cost of more false alarms. For regulatory screening this is the correct operating point: a false alarm costs an unnecessary test, whereas a missed hazard can cost lives.
3. Treat low-confidence predictions as requiring testing. Any compound whose predicted probability falls between 0.4 and 0.6 for a hazard class should be flagged for laboratory assessment rather than accepted either way.
4. Extend the training data with Malaysian regulatory sources. The model is trained on ECHA, HSDB, NITE-CMC and Safe Work Australia data. Incorporating Malaysian CLASS Regulations classifications would improve coverage of locally significant chemicals, particularly palm-oil and rubber-processing auxiliaries.
5. Inorganic and organometallic compounds are the known weak point. The descriptors used here are designed for organic molecules. Predictions for simple inorganic salts, metal oxides and mineral acids should be treated with particular caution.

6. Limitations

Compounds with uncertain predictions. 23 of 44 compounds produced at least one hazard probability in the ambiguous 0.4-0.6 band: hexane, bleaching earth, citric acid, glycerol, sulfur, mercaptobenzothiazole, acetic acid, tetramethylthiuram disulfide, xylene, acetone, cyclohexane, hydrofluoric acid, acetone, isopropanol, phosphine, tetramethylammonium hydroxide, acrolein, xylene and others.

Structural coverage. The molecular descriptors used here describe covalent organic structures. Ionic solids, elemental materials and coordination compounds are represented only approximately, so several entries in the palm-oil and rubber sectors (sodium hydroxide, zinc oxide, sulfur, graphite) sit outside the model's reliable applicability domain.

Ground truth. Performance can only be measured for compounds that already carry a GHS classification in PubChem. Compounds without one were predicted but could not be scored.

DISCLAIMER. This prediction is a computational screening tool and does not replace laboratory testing or regulatory assessment under Malaysia's Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013 (CLASS Regulations).